

What is a Senior Software Engineer at Scaleups?

How the role is typically defined at medium and large-sized venture-funded companies. Details from Honeycomb, Wise, Miro, Travelperk, Sentry, Sourcegraph, Remote.com and Urban Sports Club



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What is a senior engineer, and what are typical expectations at the senior level?

In a previous issue, we covered [what senior engineer expectations are at Big Tech companies](#). Today, we delve into how these expectations line up at scaleups, by which I mean venture-funded startups in the latter stages of funding, typically beyond Series C. Such companies usually employ more than 100 staff and often over 50 engineers.

In today's issue, we cover:

1. **Honeycomb's seniority framework.** Visualizing the increasing time between promotions, and the increasing ownership and scope of each level.
2. **Larger scaleups.** Companies with more than 1,000 employees, like Wise, Miro, and TravelPerk.
3. **Mid-size scaleups.** Scaleups with 200-1,000 employees, including Urban Sports Club, Sentry, Sourcegraph and Remote.com.
4. **Comparison to Big Tech.** Similarities exist, but the definition of "senior" differs. For example, scaleups often expect seniors to own business-critical projects, while Big Tech focuses more on maintainability.

1. Honeycomb's seniority framework

[Honeycomb](#) is an observability platform which has raised around \$150M in venture funding since being founded in 2016. The company is thoughtful about engineering leadership practices; last year, VP of engineering Emily Nakashima, shared [An Engineering Manager's Bill of Rights \(and Responsibilities\)](#), codifying cultural changes in engineering management roles. The goal was to protect engineering managers and support them in the challenges of their role.

Also last year, [Ben Darfler](#), director of engineering at Honeycomb, [shared](#) the company's approach to engineering levels – and the details show rare thoughtfulness.

The company chose not to define its engineering career ladder via a top-down process, and instead invited all engineers to weigh in. Based on common themes which emerged, Honeycomb's leadership created a new structure based on a "Problem, Solution, How and Execution" (PHSE) framework. The approach was distributed internally as a request for comment, some final adjustments were made, and the company then adopted the new ladder. It has these levels:

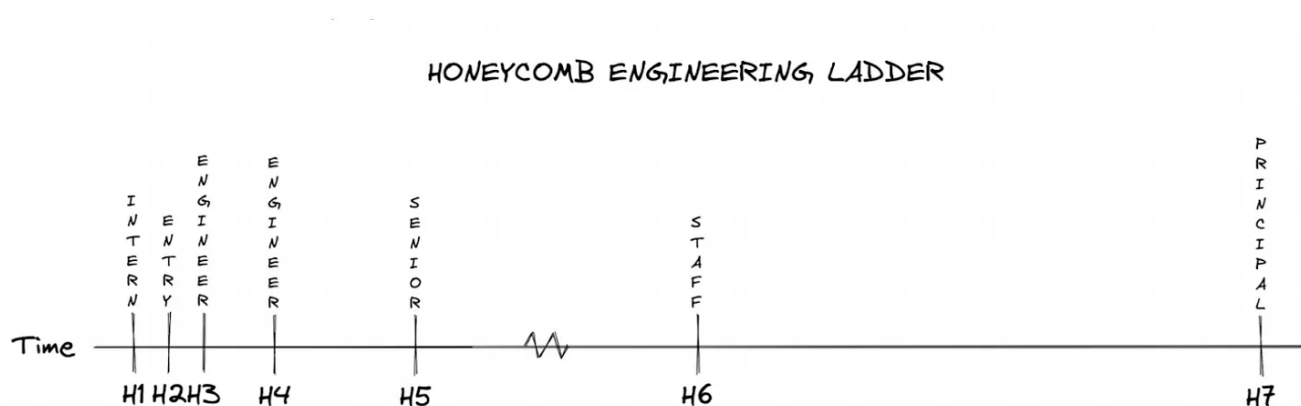
Level	Title
H1	Intern
H2	Entry-level engineer
H3	Engineer
H4	Engineer
H5	Senior engineer
H6	Staff engineer
H7	Principal engineer

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Engineering levels at Honeycomb

Honeycomb produced two diagrams illustrating important concepts:

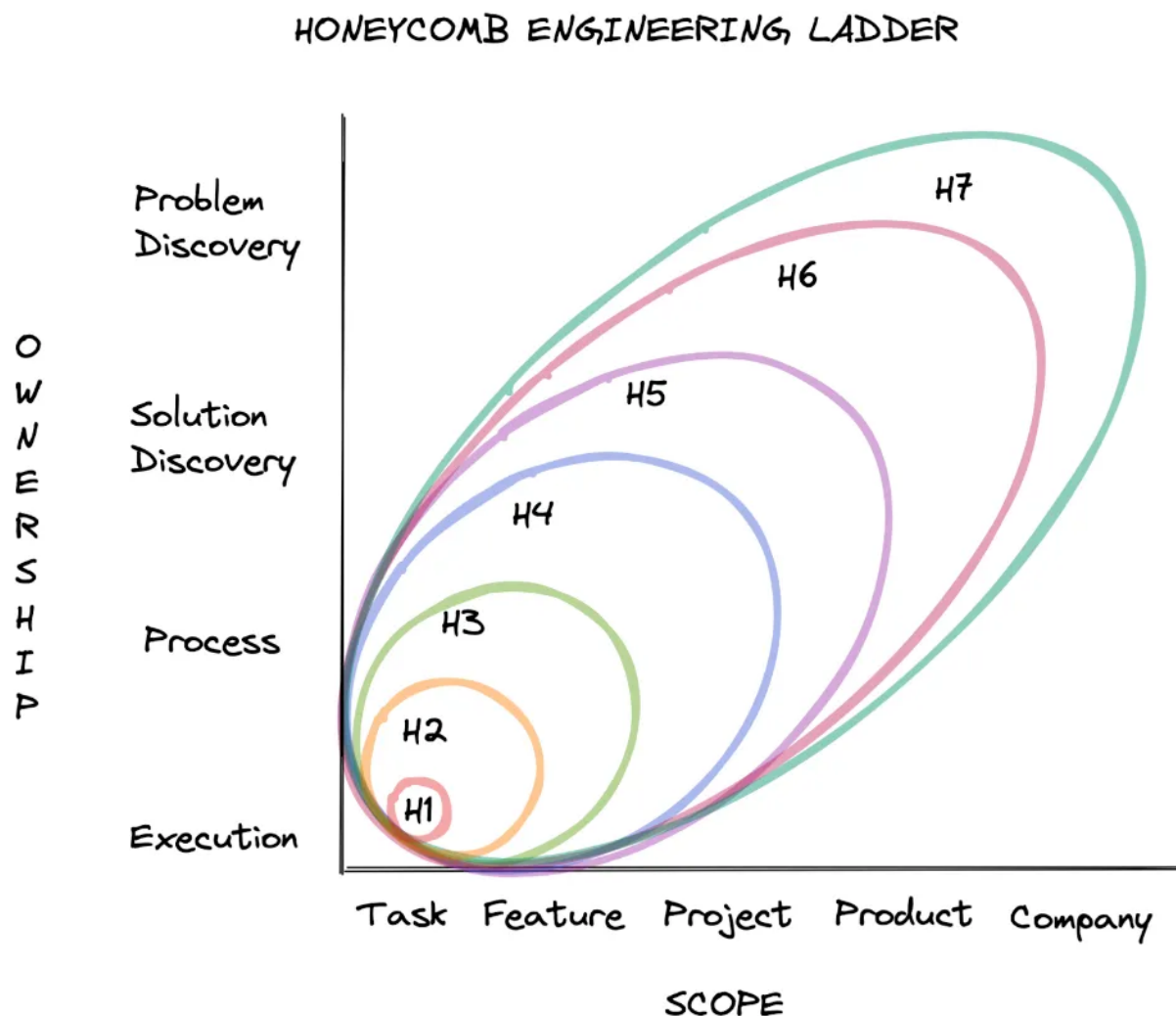
1. Length of time between levels. The higher an individual contributor (IC) goes up the ladder, the longer it typically takes to reach the next level. This differs from lower down the ladder, where promotions tend to happen faster:



Typical time to move up a level. The mark between levels H5 and H6 indicates H5 is a terminal level, beyond which there's no expectation to get promoted. Source: [Honeycomb blog](#).

This diagram is important because a "promotion velocity" like this is standard across much of the tech industry. However, many engineers don't realize the pace of promotions slows *drastically* after the senior engineer level. This can cause frustration about how long it takes to reach the next level. But it's meant to be challenging: at most tech companies the senior engineering level is considered a terminal one!

2. A growth framework, based on scope and ownership. Honeycomb produced a neat visualization of how each level brings increased ownership and scope, which incorporates previous levels.



How the scope and ownership of each level increases. This graphic depicts growing expectations and the fact that each level encompasses the expectations of preceding levels. Source: [Honeycomb blog](#).

This graph captures in a clear way the standard expectation that engineering levels contain preceding levels' scopes and ownerships. Still, it's rare to see these expectations illustrated so clearly as Honeycomb has. In my experience, these two graphs are almost universally applicable, especially at larger tech companies. *At fast-growing startups, promotions may still be quicker for early-career employees – assuming the company grows rapidly.*

So how does Honeycomb define their senior engineer level? I asked director of engineering Ben Darfler – the author of the above blog post – and he summarized the Senior expectations at Honeycomb like this:

“The Senior level is for experienced technical leaders with a broad range of practical engineering experience spanning many years. They don't just execute on their own projects; they mentor others into newly-minted senior engineers, communicate effectively cross-functionally and with those outside the organization, and are relentless in connecting technical work to business impact.

At the Senior level, engineers help set both technical and product direction.

They do this through:

- Product problem discovery: identifying technical challenges in their area of expertise
- Project solution discovery: by partnering with Product Management to design technical solutions to product problems

Senior engineers are seen as models of our core values that others can learn from.”

2. Larger scaleups

I'm defining larger scaleups as companies employing at least 1,000 people – and typically, hundreds of engineers.

Wise

[Wise](#) – formerly Transferwise – is a publicly traded Fintech, valued at \$6.5B at time of publication. The company employs more than 5,000 people in 17 countries. Its biggest engineering hubs are London (UK,) Estonia, Austin (US,) Budapest (Hungary,) and Singapore. The company not only makes its engineering levels [public](#), but also its compensation ranges across all regions. Here's how levels and compensation line up:

Level	Compensation (annual)				
	US	UK	Estonia	Singapore (SGD)	Hungary (monthly, HUF)
L1 - Junior Engineer	\$115-145K	£45-60K	€35-46K	\$81-108K	800K-1.1M
L2 - Software Engineer	\$145-180K	£60-80K	€46-64K	\$108-147K	1.1-1.5M
L3 - Senior Engineer	\$170-260K	£80-115K	€64-88K	\$147-204K	1.5-2.2M
L4 - Staff Engineer	\$240-300K	£110-150K	€88-105K	\$204-276K	2.1-2.5M
L5 - Principal Engineer	\$300-400K	£140-180K	€98-120K	\$258-330K	2.3-2.9M

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Career levels and compensation targets at Wise. These numbers are gross the gross base salary ones: equity comes on top of these. Source:

[Wise engineering career map](#)

Harsh Sinha, the CTO of Wise added additional context to the above numbers:

- The numbers refer to gross base salary.
- Equity is paid on top of these, and Wise does not have cash bonuses. The bonuses are 'loaded in' to the equity awards.
- The US numbers are clearly higher than other regions. As context, the numbers reflect what the company has found it takes to close great engineering talent in the respective markets.
- Wise has a much smaller US footprint, engineering-wise, and has scaled their engineering team heavily in the EU and UK, over the years.

Summarizing L3 (senior engineer) expectations at Wise:

Area	Typical expectation
One-liner	An engineer who delivers product-level vision, and who's expert in one or more technical or business domains
Scope	Product-level
Influence	Team level Collaborates across teams and functions like product, analytics and design
Ownership	Regularly delivers large features and product improvements of meaningful impact for Wise's product and customers
Time horizon	2-4 months
Autonomy	Autonomous in approach and may direct or coach less experienced engineers
Complexity of work	Medium to large-sized projects of 8-16 engineering weeks
Nature of work	Architects and designs services and systems using design patterns for iterative delivery and future scaling
Engineering practices	Prioritises and values undesirable/unowned work that helps the team move faster. Proactively identifies and tackles mounting technical debt through planning work and aligning the team.

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The senior level at Wise. [View this image as text here](#)

I asked [Balazs Barna](#), the Head of US Engineering at Wise to give his two cents on the senior role. He said:

"The Senior Engineer role at Wise is the equivalent to an Engineering Manager 1 on the IC track. Senior Engineers are not only expected to lead significant projects in

their teams, but they have a say in whether that feature is worth building or not. After all, there is nothing worse than an engineer solving a problem that doesn't need to be solved in the first place!

We put a lot of emphasis on communication and prioritization and the ability to unblock yourself or your team – this comes on top of the programming and design skills.

A 'typical' Senior Engineer to me is someone who seeks out difficult problems, and has the skills to solve them in a sustainable way. They strive to build things fast, but also know when we have to slow down to go faster."

Miro

[Miro](#) is a popular digital whiteboard tool, last valued at \$17.5B in 2022, which [boasts](#) NBA champion Stephen Curry among its investors. The company employs around 1,800 people across 12 regional hubs, at time of publication. Its largest engineering hubs are Amsterdam, Yerevan (Armenia), with remote positions across Europe. Miro defines engineering levels like this:

- L4: entry-level
- L5: mid-level
- L6: senior
- L7: staff
- L8: senior staff

A summary of the L6 level:

Area	Typical expectation
One-liner	Drives team-critical projects on their own and aligns stakeholders
Scope	Team
Influence	Team (usually around 5 engineers) Influence less senior folks and mentor them through code reviews, pair programming, etc.
Ownership	Project
Time horizon	Months
Autonomy	Works autonomously on team-level projects
Complexity of work	Breaks down high-level user stories into technical tasks. Takes loosely defined product features, scopes and builds them, sometimes working across teams
Nature of work	Implements system-critical components, introduces and gets buy-in for new tools, drives technical improvements that don't span more than one domain. Collaborates directly with the PM (and sometimes PMs from other teams) and helps them refine product requirements and decompose tasks
Engineering practices	Knows which tools they use well and makes effective tradeoffs when choosing them. Works closely with staff engineers to implement higher-level architecture

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The senior level at Miro. [View this image as text here](#)

L6s are not expected to drive complex projects involving multiple teams. Of course, it's a bonus when they do, and this is often seen as "exceeding expectations" at the calibrations.

As context, an engineering manager at Miro shared with me that a lot of technology changes are happening at Miro, meaning L6 engineers encounter novel new challenges. For example, they might need to adapt their services to work with a new continuous integration (CI) pipeline being rolled out company-wide, or onboard to the [Backstage developer portal](#). L6 engineers are expected to solve such challenges when their scope is manageable, or get support from more experienced engineers.

As additional detail, Miro defines the competencies by which it evaluates engineers at all levels. These are:

- Building at quality
- Critical thinking, problem solving
- Design (software)
- Leadership
- Communication and presentation
- Interviewing and hiring
- Coaching, mentoring, onboarding

TravelPerk

[Travelperk](#) is a B2B travel and expense management company, to which I'm [an advisor](#). The business employs nearly 1,000 staff, of which around 300 work in tech. Its largest engineering hubs are Barcelona (Spain,) and London (UK.) Expectations for senior software engineers can be summarized like this:

Area	Typical expectation
One-liner	Leads delivery of solutions for a wide range of problems, which may span across multiple domains, either within or without their squad (team)
Scope	Squad
Influence	Beyond their squad, as needed
Ownership	Involved in the most complex initiatives that span the whole organisation
Time horizon	Months
Autonomy	Operates with a high degree of autonomy, particularly for domains owned by their squad
Complexity of work	Leads delivery of solutions to a wide range of problems, of mid-to-high complexity. They anticipate risks, define technical solutions, and delegate tasks effectively. They have a knack for solving problems spanning multiple domains, within or beyond their squad
Nature of work	Designs, plans, and executes solutions involving multiple domains, or which impact an entire discipline. Leads improvements to existing processes, proposes beneficial changes, and ensures everyone is involved
Engineering practices	Promotes excellent quality work and adoption of best practices. Oversees multiple domains within a tribe, ensuring the implementation and accuracy of Service Level Objectives (SLOs), as well as the effectiveness of monitoring practices. Responsible for the systemic and root-cause solution to production issues

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The senior level at TravelPerk. [View this image as text here.](#)

Travelperk lists the parameters of expectations for each level:

1. Technical leadership
2. Autonomy and impact
3. Quality advocacy and monitoring
4. Process improvement and mentorship
5. Communication and collaboration
6. Talent acquisition and development

3. Mid-sized scaleups

I refer to mid-sized startups as companies employing between 200-1,000 staff.

Urban Sports Club

[Urban Sports Club](#) is a subscription-based business for sports and wellness activities at facilities across Europe. The company has more than 500 employees and raised €80M in 2021.

The scaleup [published](#) its engineering career ladder. Expectations for the senior engineer level look like this:

Area	Typical expectation
One-liner	Leads major engineering initiatives with little to no guidance
Scope	Team
Influence	Team, and sometimes across teams as well
Ownership	Team-level initiatives
Autonomy	Needs little to no guidance to lead major initiatives. Possesses a contagious drive for forward action
Complexity of work	Leads complex and business-critical coding projects, including guiding others
Nature of work	Writes production-ready code for large, complex projects. Writes self-documenting code and improves documentation without being asked.
Engineering practices	Ensures the quality of their product and services. (The company does not have dedicated QAs.) Owns their design choices for applications. Solid understanding of infrastructure concepts for their own services and products. The infrastructure and developer experience teams empower through tools and platforms. Mentors engineers on good practices, which is a big part of improving teams

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The senior level at Urban Sports Club. [View this image as text here.](#)

[Ítalo Vietro](#), VP of Engineering at Urban Sports Club, shared extra details on which skills are deemed important at senior level:

"Soft skills are a strong requirement for us, as the cost of a hire who doesn't fit our culture is immense. We have company principles for these skills that apply to any role we hire.

Much of what a senior brings is experience in system design. One thing we struggled with was hiring people who didn't have solid system design and architectural understanding. We tailored our interviews so we could hire people with that expectation. Internally, we also upskilled many engineers to get to a level that would take us forward."

One thing that surprises me is how much Urban Sports Club's expectations focused on security practices. Ítalo said this:

"In the past, we didn't have a dedicated security team, which meant both the teams and myself took on this responsibility. We later introduced practices such as security threat modeling and cloud security concepts. Teams owning all security responsibilities wasn't sustainable long term, as security requires dedicated focus. So we hired a dedicated security team."

Sentry

[Sentry](#) is a logging and performance monitoring tool, employing about 200 staff. I asked cofounder and CTO, [David Cramer](#), how the senior level is defined. He said:

"Sentry has roughly three classifications for IC engineers: software engineer, senior software engineer, and staff software engineer.

The Software Engineer role is primarily focused on getting you to a functional place within the organization. Like a lot of the industry, "Senior" has become a bit less aggressive in terms of academic background, domain knowledge, and generally on-job experience. It's pretty much expected that if you're at Sentry for 3 years, you should need to operate at our Senior level to be successful.

This expectation is partly *because* our engineering track is more lenient about achieving Senior status. Generally speaking, we'd say a Senior has 5+ years of progressive domain experience, and "Staff" is closer to the 10 year mark. Staff is

fairly similar to what you'd find at many Big Tech companies in terms of expectations. However, given we're a smaller company of roughly 400 full time employees, it may not be as demanding in domain expertise or tenure.

The senior software engineer level expects proficiency in your craft – but not mastery of it! – along with a high level of independence and ability to push decisions forward. We put a lot of emphasis on taking the initiative at Sentry, and you'll often find people succeed here without having a strong academic background.

Instead of academic qualifications, we want them to focus on what's right for the customer, the velocity of change, participating in larger conversations and sharing their opinions. Anyone in the senior or staff role is expected to contribute to organizational, architectural, and other cross-functional needs."

Sourcegraph

[Sourcegraph](#) is a code search scaleup, employing just over 200 people, about 80 of whom are software engineers. The company raised a \$125M Series D round in July 2021, at a \$2.6B valuation.

At Sourcegraph, the senior engineer definition maps to the IC3 level and this definition is [shared publicly](#). At a glance:

Area	Typical expectation
One-liner	Independently resolves complex problems. Contributes to cross-functional projects. Trains others
Scope	Larger projects
Influence	Team and cross-functional projects. Collaborates with other teams, customers, and stakeholders
Ownership	Project and their own domain
Autonomy	Operates autonomously. Requires minimal direction. Unblocks themselves and others as needed
Complexity of work	Solves complex, loosely-defined problems. May lead larger projects.
Nature of work	End-to-end work through planning, shipping, cleanup, and maintenance. Dives into unfamiliar areas of the codebase. Estimates methodically and sets realistic deadlines
Engineering practices	Deep understanding of their team's code, debugs their team's code expertly, and has mastered their tools

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The senior level at Sourcegraph. [View this image as text here.](#)

Sourcegraph shares more details on the IC3 level [in its handbook](#).

We have covered [Sourcegraph's engineering culture](#). Check out an [interview with Steve Yegge](#), Head of Engineering at Sourcegraph.

Remote.com

[Remote.com](#) is a scaleup that helps with hiring, management and payments for international teams, by taking care of payroll. The company has less than 200 people, and introduced its current engineering career ladder when it employed around 50 people. This engineering career ladder is [shared publicly](#) and defined like this:

- IC1 - Junior engineer
- IC2 - Engineer
- IC3 - Senior engineer 1
- IC4 - Senior engineer 2
- IC5 - Staff engineer 1
- IC6 - Staff engineer 2
- IC7 - Principal engineer
- IC8 - Lead principal engineer

Here is the IC3 – Senior engineer 1 – level at a glance, with input from engineering director [Pavel Zastavitskiy](#) (thank you!):

Area	Typical expectation
One-liner	Fully independent ICs. Show full ownership within the team's scope, and work closely with the team in the end-to-end development process. Senior engineers – especially at the IC4 level, can lead projects: take a definition from the product manager, plan and coordinate the work with other engineers in their team
Scope	Team
Influence	Team-related work, from idea generation to careful execution. An expert in their domain. Also influences decisions made by other teams, and wider technical strategy through the RFC process
Ownership	Team-scoped projects, with participation in cross-team initiatives
Autonomy	Fully autonomous. Handles prioritization and execution of complex features without supervision. Knows how to get unblocked, escalates proactively when support is needed. Can fill in the gaps and clarify requirements, make and proactively communicate decisions
Complexity of work	Can take on building and owning major features of the product
Nature of work	Full product development lifecycle from ideation to building and supporting users. In addition, senior engineers maintain and improve the shared codebase through direct contributions or the RFC process. Supports less experienced colleagues
Engineering practices	Seeks other senior colleagues to discuss best approaches to technical problems. Proactively seeks new tools and frameworks that can improve the product

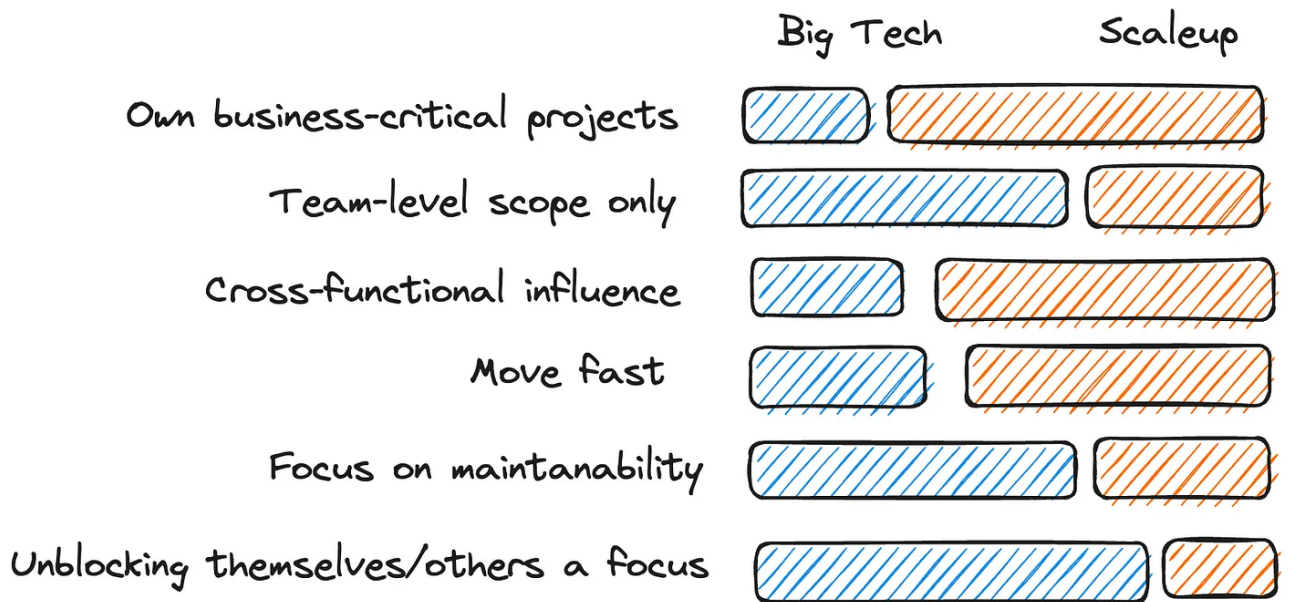
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The senior level at Remote.com. [View this image as text here.](#)

For more details, see Remote.com's [engineering career ladder](#).

4. Comparison to Big Tech

Let's try to summarize the senior engineer role, based on details from the above scaleups, and contrast it with a generalized definition of the role in Big Tech. Here's a high-level overview of the differences I noticed:



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Some of the differences in explicit expectations between the Senior level at the analyzed Big Tech companies, and the analyzed scaleups. Note that as all companies have different expectations, no such comparison will be entirely accurate.

And here is more detail on the differences. The differences are highlighted in green:

Area	Typical expectation at scaleups	Typical expectation at Big Tech
One-liner	Leads delivery of solutions for a wide range of problems with little to no guidance	Independently delivers projects within scope of the team/product
Scope	Product, larger projects or team	Team and product
Influence	Across their team Cross-functional influence	Across their team Might influence beyond the team
Ownership	Projects at team-level Participates in cross-team initiatives	Projects at team-level Services the team owns
Autonomy	Autonomous Little to no guidance required	Self-directed Delivers projects independently Unblocks themselves and their project as needed
Complexity of work	Complex, loosely defined problems. Can take on major project features and complex, business-critical projects	Moderate to high complexity within team scope. Can execute projects with other team dependencies
Nature of work	End-to-end: from ideation, planning, through shipping, cleanup and maintenance/support	From planning (sometimes even before,) to shipping and operating the software/systems
Engineering practices	Ships high quality work; mastery of their tools. Prioritizes approaches to make the team move faster. Proactive in seeking out tools and approaches to improve the main product	Improves engineering practices, product and engineering quality across the team. Considers simplicity and maintenance when designing and building software

Comparison of a 'generalized' definition of "senior engineer" across scaleups and Big Tech. [View this image as text here.](#)

In what was a bit surprising to me, the definitions at Big Tech and scaleups don't seem very different. I'll assume some of this could be due to the work being similar at Big Tech and at a scaleup, as in both environments smaller teams build products which large numbers of people use. Also, there's plenty of cross-pollination between Big Tech and scaleups, with engineers and managers moving between them.

There are some differences worth highlighting, though:

Scaleups care less about the "team" scope of seniors. At Big Tech the expected scope for seniors is almost always team level. At scaleups, this isn't always the case, especially not at smaller places, where the expectation is to get a project done, even when it involves working with or influencing stakeholders or other teams across the company. *This difference is most likely down to size: Big Tech tends to have hundreds of engineering teams, and scaleups have far fewer.*

Big Tech companies emphasize 'unblocking' much more. Being blocked by another team, and senior engineers unblocking themselves or their team, is a common expectation at Big Tech. At scaleups, this concept seems to be largely absent from explicit expectations. I assume this is because things are less formalized at scaleups, and engineers are likely more approachable.

A team at a Big Tech company might receive dozens of requests from other teams, which they don't have time to quickly deal with. Engineers need to familiarize themselves with unblocking their team in formal ways, such as via escalation. So, yes, engineers "unblock" themselves at both places, but getting unblocked is usually less challenging to do at scaleups than in Big Tech.

Scaleups expect seniors to solve much more complex problems... on paper. At Big Tech, the senior role is not expected to solve the most complex problems across the business – or even a group. Staff and principal engineer roles are expected to do this. Based on expectations within scaleups, it seems senior engineers must take on more complex projects.

What I think is happening here is that a scaleup's engineering team is much smaller, so every engineer naturally has more scope than in Big Tech. This is true for senior engineers and all other levels.

So yes, senior engineers at smaller scaleups will frequently make major changes to the flagship product and ship functionality which most customers use. At Big Tech, this is more rare, and the engineers leading such projects are likely Staff-or-above level – or if they are not, then delivering one of those projects can already serve as the solid promotion case to get to that level.

Moving faster and improving the product is more emphasized at scaleups.

Regarding engineering excellence, both scaleups and Big Tech expect quality work from senior engineers. However, speed feels more important at scaleups than at Big Tech, where shipping maintainable solutions is much more frequently emphasized – but hardly mentioned at scaleups.

This is interesting, as I see both approaches as “moving fast,” but in different environments:

- At scaleups, moving fast is, well... moving fast. Shipping quality code and doing so quickly.
- At Big Tech, moving fast means shipping maintainable solutions, and doing this quickly *enough*. Big Tech tends to have more teams working on projects, so if a solution is not maintainable, then the launch might be faster, but iterating the product will be slower. So overall, it can be a faster approach to invest a bit more effort in the initial launch and ship maintainable software. Don't forget that for experienced-enough engineers, shipping maintainable software and shipping quickly aren't mutually exclusive!

Takeaways

There are many similarities between Big Tech and tech scaleups. Both types of company tend to treat software engineering as [a profit center](#) and both care about moving fast with high quality. A big difference is the size of a company – and often its age – and also the maturity of its product offering.

We can get interesting insights into some differences between these types of companies by looking at the senior level. My sense is that at Big Tech the senior engineer role is more limited due to the complex organizational setup, which limits expectations of the impact an engineer makes *within* their team. Of course, it's a bonus when engineers deliver projects that don't fully fall inside of their team, but this isn't a baseline expectation.

At scaleups, the baseline expectation is often higher. Circling back to how Sentry cofounder David Cramer puts it:

“Anyone in the senior role is expected to contribute to organizational, architectural, and other cross-functional needs. We want [senior engineers] to focus on what is right for the customer, the velocity of change, participate in larger conversations and share their opinions.”

There's likely a lot more space for senior-level work at scaleups thanks to the business being smaller, and there often being too few people for the work at hand. For engineers in the early or mid stage of their careers, joining a scaleup could mean getting more responsibility sooner in terms of the impact their work has on the company – than at Big Tech.

If you have any observations about the senior role at scaleups, please share in the comments:

Now we've covered Big Tech and scaleups, we'll examine the senior engineer role at startups and digital agencies in a follow-up article.



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3 Comments



Write a comment...



Roman Jul 25 ❤️ Liked by Gergely Orosz

Very accurate coverage as always. I lead a few groups at a large scaleup, and can confirm that our thinking is similar.

One aspect worth mentioning is operations and design for large scale - specifically, ability to resolve operational incidents, think about non-functional requirements (resilience, compliance, latency), future scale needs etc. I suspect for Big Tech this would be front and center. For scaleup, it's also an important topic, though I typically expect that to come up stronger at Staff+ levels.

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Krisztián Kuntner Jul 27 ❤️ Liked by Gergely Orosz

Great story about product-focused companies. I am curious to see the differences in defining seniority at consulting companies, where there is less direct product-related ownership involved (or at least it is now your own product). Do you have insights on this, or plan to cover them in a subsequent issue?

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